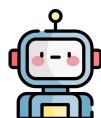


Visual Chinese Grammatical Error Correction

Track 1 Sentence Level

步带来给我挺多好处

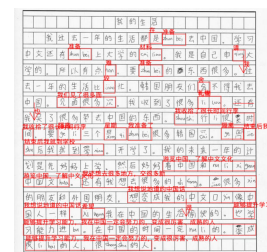


Output

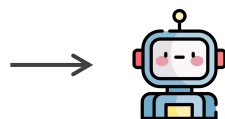
"predict_text": "跑步给我带来挺多好处",
"Edit_list":
(44,7)(120,89), '跑', Insertion
(124,5)(459,89), '给我带来',
Word Order
(519,6)(581,74), '挺', Substitute
(670,13)(732,81), '处', Substitute

Visualize

步带来给我挺多好处



Track 2 Paragraph Level



Evaluation 1

Image

步带来给我挺多好处

Source

步带来给我X多好处

Target

跑步给我带来挺多好处

Predict

这一步带来了挺多好处

Cherrant

S 步带来给我X多好处
T0-A0 跑步给我带来挺多好处
A 0 0||M||跑||REQUIRED||-NONE-||0
A 1 5||W||给我带来||REQUIRED||-NONE-||0
A 5 6||S||挺||REQUIRED||-NONE-||0
A 8 9||S||处||REQUIRED||-NONE-||0



Span-Based Correction					
TP	FP	FN	Prec	Rec	F0.5
1	2	3	0.3333	0.25	0.3125

Evaluation 2

Target

步带来给我挺多好处

Insert Word-Order Substitute Substitute
① 跑 ② 给我带来 ③ 挺 ④ 处

Predict

步带来给我挺多好处

Insert Insert Delete Insert Substitute
① 这一 ② 了 ③ ④ 挺 ⑤ 处

Iou =

Area of Overlap
Area of Union



Iou >= 0.5, box ✓
Iou < 0.5, box ✗

Target Boxes: 4
Predict Boxes: 5
Box + Text : 2
Box + Op : 2
Box + Tex + Op : 1

	Box	Correct Text	Op
Pred_e1	✓	✗	✓
Pred_e2	✗	✗	✗
Pred_e3	✓	✗	✗
Pred_e4	✓	✓	✗
Pred_e5	✓	✓	✓

Calculation:

Box + Text + Op

$$\text{Prec} = \frac{\text{Nums of [Box + Text + Op]}}{\text{Predict Boxes}}$$

$$= \frac{1}{5}$$

$$\text{Rec} = \frac{\text{Nums of [Box + Text + Op]}}{\text{Target Boxes}}$$

$$= \frac{1}{4}$$

$$\text{F0.5} = \frac{(1 + 0.5^2) * (\text{Prec} * \text{Rec})}{0.5^2 * \text{Prec} + \text{Rec}}$$